Temiloluwa Adeoti

Senior Applied AI Engineer

Senior Applied AI Engineer specializing in the architecture and operation of production AI systems. Designs data-centric machine learning platforms that integrate multimodal models, scalable cloud infrastructure, and observability frameworks to deliver reliable AI services in real-world environments. Experienced in building end-to-end AI systems—from data ingestion pipelines and model adaptation to deployment, monitoring, and lifecycle management. Combines deep machine learning expertise with cloud-native engineering practices to operationalize AI safely and cost-effectively at scale. Background spanning industrial field engineering, enterprise technology consulting, and applied AI engineering, providing a practical understanding of noisy real-world data and operational system constraints.

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KEY ACHIEVEMENTS

- Architected multimodal AI systems converting receipt images into structured retail intelligence.
- Built cloud-native AI platforms on AWS supporting scalable inference and automated retail data ingestion.
- Implemented model observability and evaluation frameworks improving reliability in noisy real-world data environments.
- Delivered enterprise AI and data platform solutions for Allianz, Volkswagen, and Porsche.

TECHNICAL LEADERSHIP AND ARCHITECTURE

AI & Machine Learning: Python, PyTorch, Hugging Face Ecosystem, LoRA / PEFT, Retrieval-Augmented Generation, Multimodal Processing, Model Evaluation Pipelines

AI Systems & Infrastructure: Kubernetes, Docker, Terraform, GitLab CI/CD, Model Serving Architectures

Cloud Platforms: AWS (Bedrock, SageMaker, Lambda, SQS, EMR), Azure (Synapse, Azure OpenAI)

Observability & Operations: Prometheus, Grafana, ELK Stack, Production Monitoring, Drift Detection, AI System Observability

Data Engineering: PySpark, SQL, Data Pipeline Architecture

Systems Languages: Rust (systems experimentation), Go (Kubernetes ecosystem)

ENGINEERING EXPERIENCE

Machine Learning Engineer
July 2024 - Present

YouGov (formerly GfK), Nuremberg, Germany

Design and operation of production AI systems for retail intelligence and multimodal document processing.

- Architected and deployed a production Receipt Intelligence Platform combining OCR, vision-language models, and LLM pipelines to extract structured FMCG retail data across international receipt formats.
- Owned the end-to-end lifecycle of AI workflows including ingestion pipelines, model inference systems, evaluation frameworks, and monitoring infrastructure.
- Designed data-centric evaluation pipelines including synthetic dataset generation and error-driven feedback loops to improve model robustness against noisy real-world inputs.
- Built scalable inference infrastructure using AWS Bedrock, Lambda, SQS, MongoDB, and Terraform enabling automated high-volume retail data processing.
- Implemented model observability frameworks tracking model behavior, latency, and operational regressions.
- Supported controlled model lifecycle management using SageMaker Pipelines and GitLab CI/CD.

Machine Learning Consultant
April 2022 - May 2024

Machine Learning Reply GmbH, Munich, Germany

Delivered enterprise AI and data platform solutions across finance and automotive industries.

- Delivered applied AI solutions and data platforms for enterprise clients including Allianz, Volkswagen, and Porsche.
- Built containerized RAG-based document intelligence systems on Azure enabling privacy-preserving enterprise knowledge workflows.
- Established repeatable deployment patterns for ML workloads on Kubernetes with integrated observability and monitoring.
- Migrated approximately 1 PB of legacy infrastructure from Cloudera to AWS redesigning pipelines using Spark on EMR with modern CI/CD practices.
- Designed secure enterprise ingestion pipelines integrating on-prem APIs with Azure Synapse.

Data Scientist (Research Engineer)
May 2021 - March 2022

Fraunhofer IKS, Munich, Germany

Research focused on reliable AI systems for safety-critical environments.

- Developed hybrid anomaly detection systems combining deep learning with rule-based reasoning.
- Applied contrastive learning (MoCo) to improve robustness in sparse anomaly detection scenarios.
- Collaborated with domain experts to transition research prototypes into production-ready system designs.

Field Engineer
Dec 2015 - Apr 2018

SLB (Schlumberger), Land & Offshore Operations

- Led end-to-end well intervention operations including matrix acidizing, nitrogen lift, and fishing operations for major operators including Total, Shell, and ExxonMobil.
- Installed field instrumentation and performed real-time data analysis guiding operational decisions in complex environments.
- Managed client relations, logistics, and operational financials across field operations.
- Produced technical reports and operational analyses supporting technical sales and business development.

EDUCATION

M.Sc. Data Analytics
2021
Grade 1.6

Universität Hildesheim

B.Sc. Electrical & Electronic Engineering
First Class Honours

University of Ibadan

CERTIFICATIONS

- AWS Machine Learning Specialty
- AWS Solutions Architect Associate
- Terraform Associate
- Azure Fundamentals

TEACHING & COMMUNITY

Data Analytics Instructor
2023–2025

ReDI School of Digital Integration

AWS Community Builder
2021–2025

AWS Community Builder

NLP Tutor
2021

DeepLearning.AI